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Exp 5.1

PROGRAM

```
clc; clear; close all;

%%% --- Input Parameters ---

fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%%% --- Frequency Response Plots ---

figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');

figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');

figure;
```

```
freqz(b_hann,1);
```

```
title('High-pass FIR using Hanning Window');
```

OUTPUT

--- High-Pass Coefficients (Rectangular Window) ---

-0.0089

0

0.0097

0.0143

0.0106

0

-0.0117

-0.0175

-0.0131

0

0.0148

0.0225

0.0171

0

-0.0202

-0.0314

-0.0247

0

0.0318

0.0524

0.0445

0

-0.0741

-0.1572

-0.2223

0.7407

-0.2223

-0.1572

-0.0741

0

0.0445

0.0524

0.0318

0

-0.0247

-0.0314

-0.0202

0

0.0171

0.0225
0.0148
0
-0.0131
-0.0175
-0.0117
0
0.0106
0.0143
0.0097
0
-0.0089
--- High-Pass Coefficients (Hamming Window) ---
-0.0007
0
0.0009
0.0016
0.0015
0
-0.0024
-0.0044
-0.0039
0
0.0060
0.0103
0.0088
0
-0.0128
-0.0217
-0.0184
0
0.0268
0.0464
0.0410
0
-0.0725
-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
0
0.0410

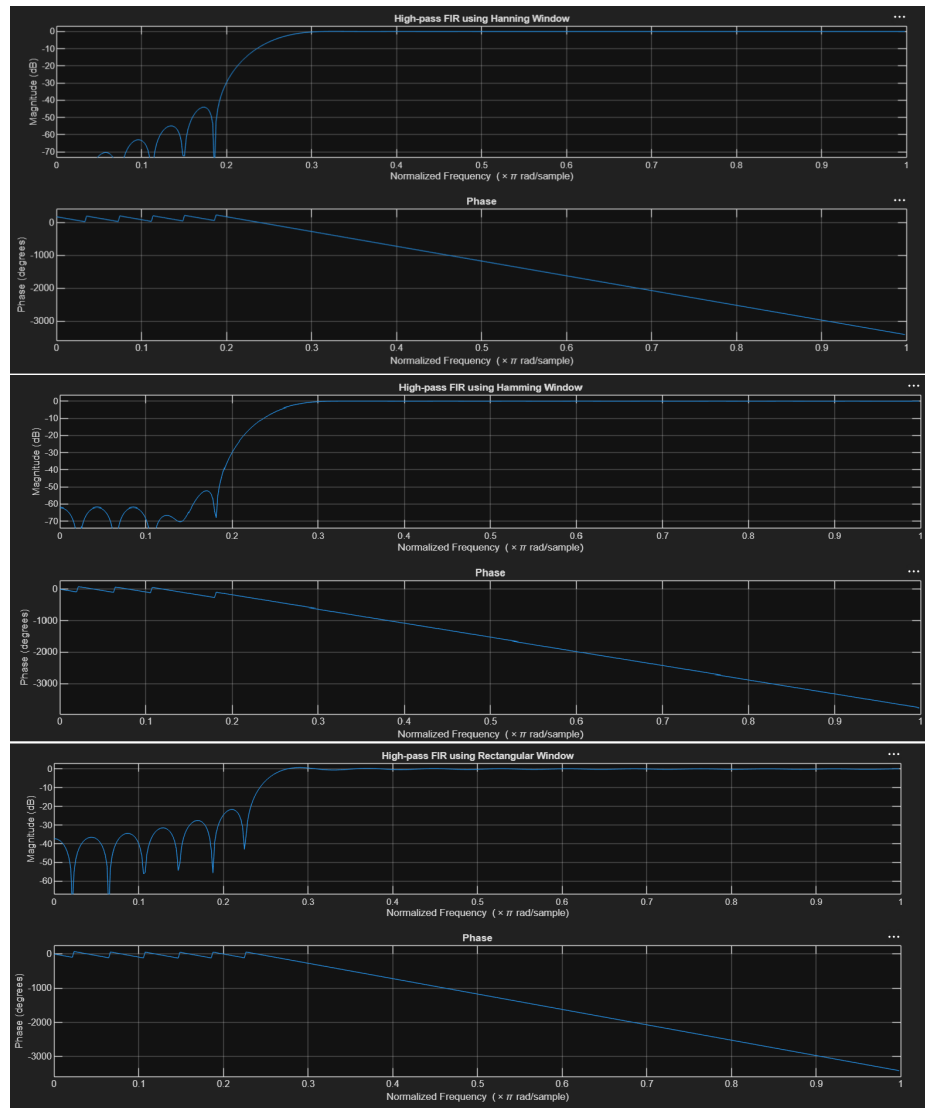
0.0464
0.0268
0
-0.0184
-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044

-0.0024
0
0.0015
0.0016
0.0009
0
-0.0007

--- High-Pass Coefficients (Hanning Window) ---

-0.0000
0
0.0003
0.0008
0.0009
0
-0.0020
-0.0038
-0.0035
0
0.0057
0.0100
0.0087
0
-0.0127
-0.0216
-0.0183
0
0.0267
0.0464
0.0410
0
-0.0726
-0.1568

0.2243
0.7500
-0.2243
-0.1568
-0.0726
0
0.0410
0.0464
0.0267
0
-0.0183
-0.0216
-0.0127
0
0.0087
0.0100
0.0057
0
-0.0035
-0.0038
-0.0020
0
0.0009
0.0008
0.0003
0
-0.0000



Ex:5.2

PROGRAM

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000;    % High-pass cutoff frequency (Hz)

fs = 8000;    % Sampling frequency (Hz)

N = 50;       % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm);
```

```
%% --- Frequency Response Plots ---
```

```
figure;
```

```
freqz(b_hamm,1);
```

```
title('High-pass FIR using Hamming Window');
```

```
title('High-pass FIR using Hanning Window');
```

OUTPUT

```
--- High-Pass Coefficients (Hamming Window) ---
```

```
-0.0007
```

```
0
```

```
0.0009
```

```
0.0016
```

```
0.0015
```

```
0
```

```
-0.0024
```

```
-0.0044
```

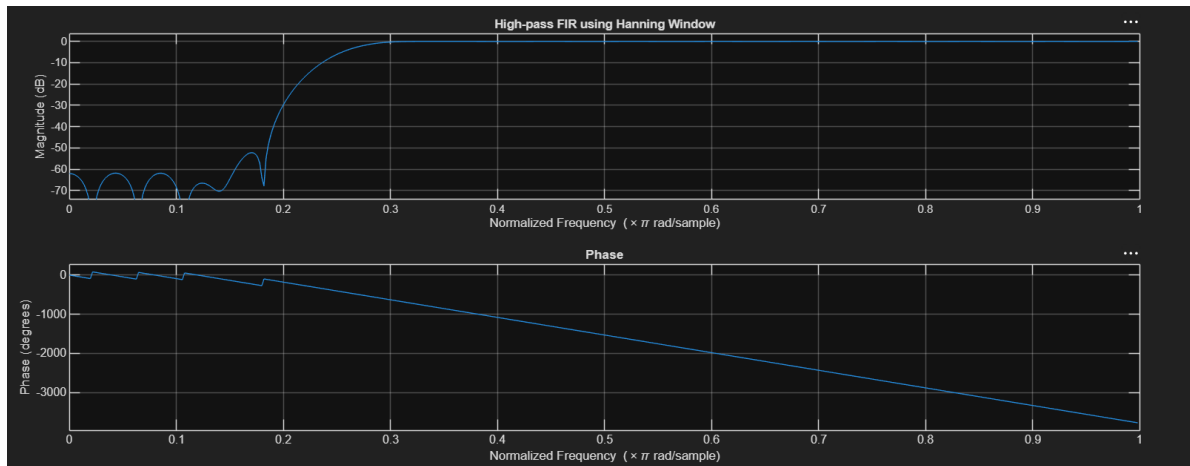
```
-0.0039
```

```
0
```

```
0.0060
```

```
0.0103
```

0.0088
0
-0.0128
-0.0217
-0.0184
0
0.0268
0.0464
0.0410
0
-0.0725
-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
0
0.0410
0.0464
0.0268
0
-0.0184
-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044
-0.0024
0
0.0015
0.0016
0.0009
0
-0.0007



Ex. : 5.3

PROGRAM

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000;    % High-pass cutoff frequency (Hz)
fs = 8000;    % Sampling frequency (Hz)
N = 50;       % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');

%% --- Frequency Response Plots ---

figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

OUTPUT

```
--- High-Pass Coefficients (Hanning Window) ---
-0.0000
    0
 0.0003
 0.0008
 0.0009
```

0
-0.0020
-0.0038
-0.0035
0
0.0057
0.0100
0.0087
0
-0.0127
-0.0216
-0.0183
0
0.0267
0.0464
0.0410
0
-0.0726
-0.1568
-0.2243
0.7500
-0.2243
-0.1568
-0.0726
0
0.0410
0.0464
0.0267
0
-0.0183
-0.0216
-0.0127
0
0.0087
0.0100
0.0057
0
-0.0035
-0.0038
-0.0020
0
0.0009
0.0008
0.0003
0
-0.0000

